Brian Wonjun Lee

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Education

University of Pennsylvania — Philadelphia, PA
MSE in Computer Graphics & Game Technology (CGGT), Submatriculation — *Expected May 2028*
BA in Computer Science & Design (Double Major), Benjamin Franklin Scholars — *Expected May 2027*
Relevant Coursework: Interactive Computer Graphics, Computer Animation, 3-D Computer Modeling, Introduction to Algorithms, Introduction to Computer Systems, Programming Languages & Techniques

Experience

Software Engineer Intern, Aleph Lab (Y Combinator) — *June 2026 – Aug 2026 · San Francisco (Remote)*

- Delivered an app-wide **React Native** redesign from Figma to production, sequencing information architecture ahead of the visual system so users kept their bearings, then carrying a **33-section** design system onto every screen
- **Built the company's lifecycle notification system end to end, then made it autonomous** — per-stage copy, anti-nag back-off and a 20% measurement holdout, clearing its own safety guardrails at **91% device reach**; widening eligibility to every account role reached **~2.7×** more families and produced its first paid conversion
- **Packaged the in-game AI character as a versioned SDK outside studios can build on** — typed APIs, per-session isolation and capability enforcement at a boundary the creator cannot cross, **136 tests** — validated by an external builder shipping a game mode without access to our repository
- **Made a children's app safe from notification injection** — anyone with a user ID could push arbitrary content until I bearer-authed four public endpoints, **zero caller changes**

Software Engineer Intern, BitMango (Puzzle1 Studio) — *June 2025 – Aug 2025 · Pangyo, South Korea*

- Maintained a live portfolio of **50+ Unity mobile game titles**, improving UI/UX and core gameplay stability across releases
- Resolved **100+ QA-reported bugs** by modifying Unity prefabs and C# scripts, shipping fixes in production releases
- Upgraded SDKs and platform modules fleet-wide to meet app-store requirements and sustain availability

Project Lead & Product Designer, Penn Spark — *Jan 2025 – Present · Philadelphia, PA*

- Led a cross-functional team of **15+** designers and developers to ship client-facing **0-to-1 products**
- Defined end-to-end user flows, interaction patterns, and reusable interface systems across multiple deliverables
- Shipped polished, user-centered features across semester-long client project cycles

Military Interpreter, Capital Defense Command (ROK Army) — *Dec 2022 – June 2024 · Seoul, South Korea*

- Provided real-time English–Korean interpretation in high-security, multi-agency environments for international stakeholders
- Translated sensitive documents and briefings with precision, enabling coordination across government and defense teams

Database Engineer Intern, IT-Farm — *June 2022 – Aug 2022 · Seongnam, South Korea*

- Designed and implemented a SQL database prototype managing **1M+ data entities** using Oracle and SQLite
- Processed and organized semiconductor datasets from **20+ client companies**, including Samsung, SK, and LG
- Wrote and optimized SQL queries to improve retrieval and management efficiency across large-scale datasets

Projects

Mini Minecraft | *C++, OpenGL, GLSL* — *Apr 2026*

- **Made a from-scratch voxel world read as varied rather than noisy** — **7-biome procedural generation** (Perlin/FBM/Voronoi) and **3D Perlin caves** in a C++/OpenGL engine (team of 3)
- **Made that world legible** — the GLSL pipeline I owned: PCF shadows, screen-space reflections, vertex AO, day-night cycle

Passenger | *Unreal Engine 5, HLSL, Maya* — *Mar 2026 – Apr 2026*

- **Gave a short film its entire visual identity** — 2 **HLSL post-process shaders** (anisotropic Kuwahara, cell shader) over GBuffer channels, delivered as a ray-traced 2560×1080 render
- **Brought its character to life** — rigged and animated in Maya (IK limbs, spline-IK spine) driving an Unreal Animation Blueprint

Mini Maya | *C++, OpenGL, Qt* — *Feb 2026 – Mar 2026*

- **Made arbitrary meshes editable in place** — a **half-edge structure** giving topology-aware traversal, plus an OBJ parser accepting arbitrary n-gons, rendered via OpenGL VBOs
- **Turned coarse geometry into smooth surfaces** with **Catmull–Clark subdivision** and topology ops in a Qt editor

Technical Skills

Game Engines: Unity, Unreal Engine 5, C#, C++, Animation Blueprints, prefab & gameplay systems
Graphics & 3D: OpenGL, GLSL, HLSL, Maya, Blender, Substance Painter
Languages: C++, C#, Python, TypeScript, JavaScript, Java, Kotlin, SQL
Frameworks, Infra & AI: React Three Fiber, React Native (Expo), Minecraft modding, Git, Docker, Claude Code, Codex, MCP
Practices: Trunk-based development, feature flags & staged rollout, CI/CD, code review, unit & regression testing